

Module 3 – Advanced Prompting Techniques

Demo 3 – Auto-Prompt Generation

Objective:

To demonstrate how to auto-generate effective, tailored prompts using only a task description, examples, or meta-prompts, with iterative refinement and variation, without writing prompts manually.

Problem Statement

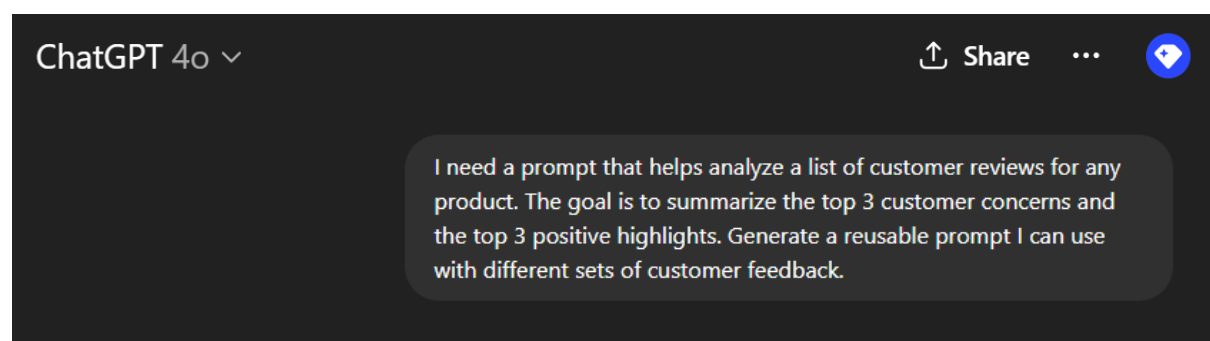
Company: TrendHive, a fast-growing online fashion retailer.

Problem: The insights team is overwhelmed by the need to analyze and summarize large volumes of customer feedback across multiple product lines.

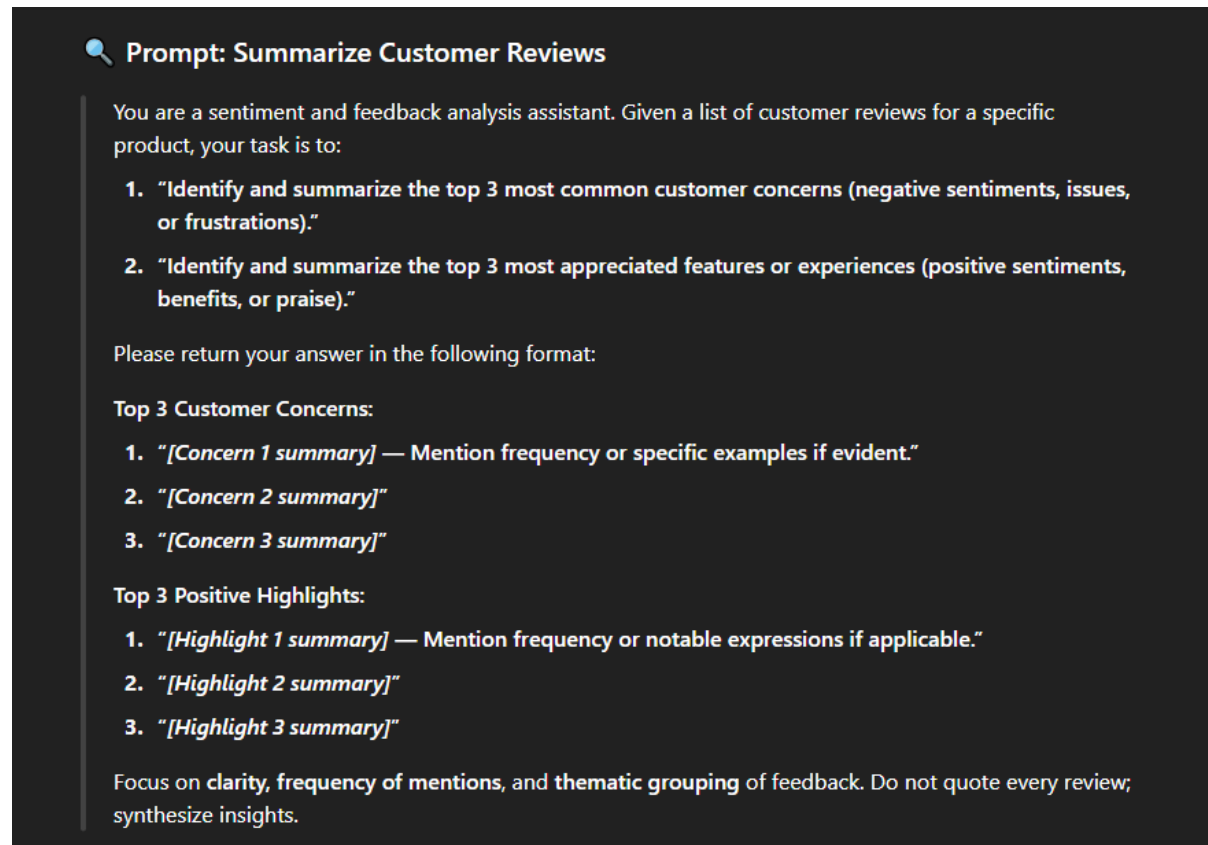
Need: Create a prompt that enables the team to automatically summarize key customer concerns and suggestions from raw feedback, ideally, the prompt should be reusable across categories and scalable.

Task Description → Prompt Generation

Prompt: I need a prompt that helps analyze a list of customer reviews for any product. The goal is to summarize the top 3 customer concerns and the top 3 positive highlights. Generate a reusable prompt I can use with different sets of customer feedback.



ChatGPT Response:



Prompt: Summarize Customer Reviews

You are a sentiment and feedback analysis assistant. Given a list of customer reviews for a specific product, your task is to:

1. "Identify and summarize the top 3 most common customer concerns (negative sentiments, issues, or frustrations)."
2. "Identify and summarize the top 3 most appreciated features or experiences (positive sentiments, benefits, or praise)."

Please return your answer in the following format:

Top 3 Customer Concerns:

1. "[Concern 1 summary] — Mention frequency or specific examples if evident."
2. "[Concern 2 summary]"
3. "[Concern 3 summary]"

Top 3 Positive Highlights:

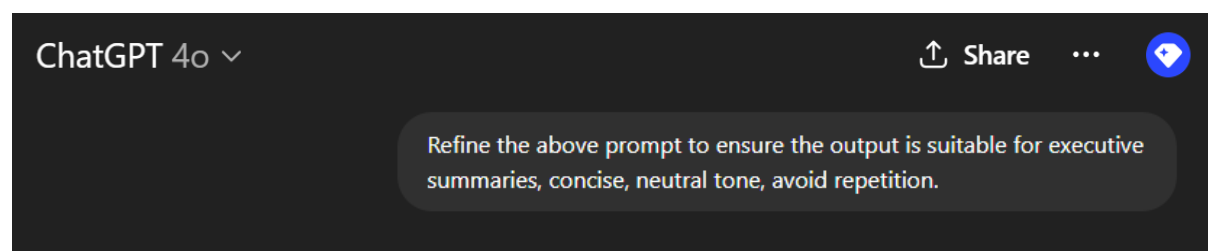
1. "[Highlight 1 summary] — Mention frequency or notable expressions if applicable."
2. "[Highlight 2 summary]"
3. "[Highlight 3 summary]"

Focus on clarity, frequency of mentions, and thematic grouping of feedback. Do not quote every review; synthesize insights.

Purpose: This is **prompt bootstrapping** from just a task description.

Refining Prompt to add Business Constraints

Prompt: Refine the above prompt to ensure the output is suitable for executive summaries, using a concise and neutral tone, and avoiding repetition.



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Refine the above prompt to ensure the output is suitable for executive summaries, concise, neutral tone, avoid repetition.

ChatGPT Response:

Here's the **refined** version of the prompt, tailored to produce **executive-friendly summaries** with a **concise, neutral tone** and **minimal redundancy**:

Prompt: Executive Summary of Customer Reviews

You are a customer insights analyst. Analyze the following list of customer reviews for a specific product. Your objective is to extract actionable insights in a format suitable for an executive summary.

Please provide:

Key Customer Concerns (Top 3):

- "Summarize the three most frequently mentioned issues or negative themes."
- "Use clear, concise phrasing."
- "Avoid repetition; group similar points under unified themes."

Key Positive Highlights (Top 3):

- "Summarize the three most consistently praised features or experiences."
- "Maintain a neutral, business-appropriate tone."
- "Focus on points that suggest product strengths or user satisfaction."

Guidelines:

- "Avoid quoting individual reviews."
- "Prioritize clarity, frequency, and thematic grouping over exact wording."
- "Format output as a brief bullet list suitable for senior stakeholder consumption."
- "Do not speculate or exaggerate—focus only on what is clearly supported by the data."

Example Output Format:

Key Customer Concerns:

- Difficulty with initial setup and onboarding process.
- Performance issues during peak usage times.
- Limited customization options compared to competitors.

Key Positive Highlights:

- Responsive and helpful customer support.
- Intuitive user interface improves productivity.
- Reliable core functionality across standard use cases.

Few-Shot Inference → Prompt Pattern Extraction

Prompt:

Below are two examples of customer feedback and their summaries. Based on these, generate a prompt that I can reuse to produce similar summaries for other datasets.

Input-Output Pair 1

Customer Reviews (Input):

"The shoes look great but started tearing within a week."

"Fast delivery, but the quality isn't what I expected."

"I love the design and color options!"

"Size runs smaller than advertised. Had to return it."

"Customer support was helpful during the exchange process."

"Material feels cheap and uncomfortable."

Summary (Output):

Customer Concerns:

Poor durability and product quality

Incorrect sizing issues

Material not meeting expectations

Customer Praise:

Appealing design and color variety

Fast delivery

Helpful customer support

Input-Output Pair 2

Customer Reviews (Input):

"The fabric is soft and feels premium."

"Shipment took over a week, very slow."

"Easy checkout process."

"The product looked different from the pictures."

"Love the eco-friendly packaging!"

"The item smelled weird when opened."

Summary (Output):



Customer Concerns:

Delivery delays

Product not matching images

Unpleasant odor on arrival

Customer Praise:

Premium-feel fabric

Smooth checkout experience

Environmentally friendly packaging

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"I love the design and color options!"

"Size runs smaller than advertised. Had to return it."

"Customer support was helpful during the exchange process."

"Material feels cheap and uncomfortable."

Summary (Output):**Customer Concerns:**

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Material not meeting expectations

Customer Praise:

Appealing design and color variety

Fast delivery

Helpful customer support

Input-Output Pair 2**Customer Reviews (Input):**

"The fabric is soft and feels premium."

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"Easy checkout process."

"The product looked different from the pictures."

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Summary (Output):**Customer Concerns:**

Delivery delays

Product not matching images

Unpleasant odor on arrival

Customer Praise:

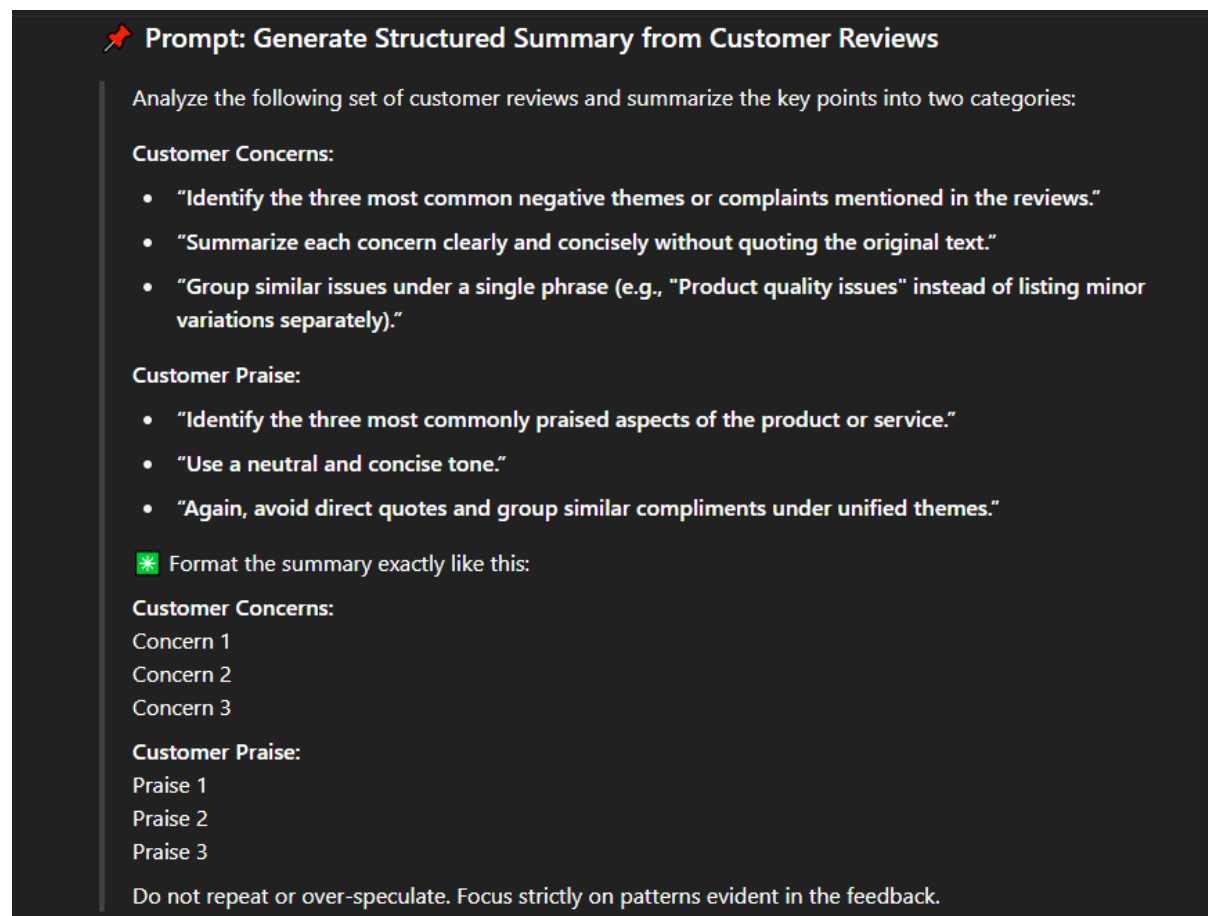
Premium-feel fabric

Smooth checkout experience

Environmentally friendly packaging



ChatGPT Response:



✦ Prompt: Generate Structured Summary from Customer Reviews

Analyze the following set of customer reviews and summarize the key points into two categories:

Customer Concerns:

- "Identify the three most common negative themes or complaints mentioned in the reviews."
- "Summarize each concern clearly and concisely without quoting the original text."
- "Group similar issues under a single phrase (e.g., "Product quality issues" instead of listing minor variations separately)."

Customer Praise:

- "Identify the three most commonly praised aspects of the product or service."
- "Use a neutral and concise tone."
- "Again, avoid direct quotes and group similar compliments under unified themes."

✳ Format the summary exactly like this:

Customer Concerns:
Concern 1
Concern 2
Concern 3

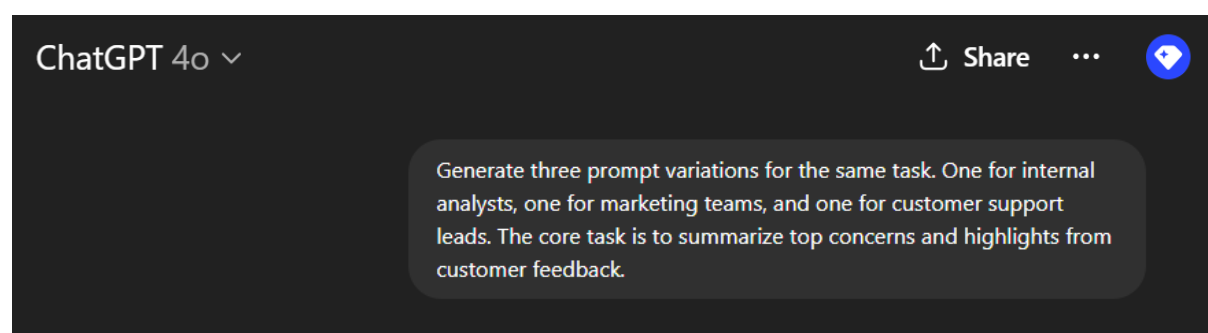
Customer Praise:
Praise 1
Praise 2
Praise 3

Do not repeat or over-speculate. Focus strictly on patterns evident in the feedback.

Purpose: Demonstrates **reverse-engineering prompts from examples**.

Tone Variation for Target Audiences

Prompt: Generate three prompt variations for the same task. One for internal analysts, one for marketing teams, and one for customer support leads. The core task is to summarize top concerns and highlights from customer feedback.



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Generate three prompt variations for the same task. One for internal analysts, one for marketing teams, and one for customer support leads. The core task is to summarize top concerns and highlights from customer feedback.

ChatGPT Response:

1. For Internal Analysts (Data-Focused, Neutral)

You are an insights analyst reviewing customer feedback. Your task is to identify key trends in the data for internal reporting.

Instructions:

- "Analyze the customer reviews provided."
- "Summarize the top 3 recurring concerns and top 3 consistently positive themes."
- "Group similar points under broader issue categories (e.g., "product quality issues" or "positive customer service experience")."
- "Maintain a neutral, data-oriented tone."
- "Avoid direct quotes or assumptions beyond the data."

Output Format:

Customer Concerns:

1. "..."
2. "..."
3. "..."

Customer Praise:

1. "..."
2. "..."
3. "..."

2. For Marketing Teams (Value-Focused, Messaging-Oriented)

You are summarizing customer reviews to extract insights useful for product positioning and promotional messaging.

Instructions:

- "Review the customer feedback below."
- "Identify the top 3 areas of praise that reinforce brand strengths or value propositions."
- "Highlight the top 3 concerns that may impact perception or conversion."
- "Phrase the summary in a concise, messaging-aware tone suitable for internal brand discussions."
- "Focus on customer sentiment, themes, and possible language to reinforce or avoid in marketing copy."

Output Format:

Positive Highlights for Messaging:

1. "..."
2. "..."
3. "..."

Perception Risks to Monitor:

1. "..."
2. "..."
3. "..."

3. For Customer Support Leads (Action-Oriented, Service-Driven)

You are reviewing customer reviews to improve support strategies and reduce repeat complaints.

Instructions:

- "Examine the feedback to identify the top 3 recurring issues or frustrations that may require support process improvements or FAQs."
- "Identify the top 3 aspects of the customer experience that are praised, which support teams should continue reinforcing."
- "Focus on service-level trends, customer experience pain points, and opportunities to improve resolution workflows."
- "Keep the tone practical and solution-focused."

Output Format:

Key Support Pain Points:

1. "..."
2. "..."
3. "..."

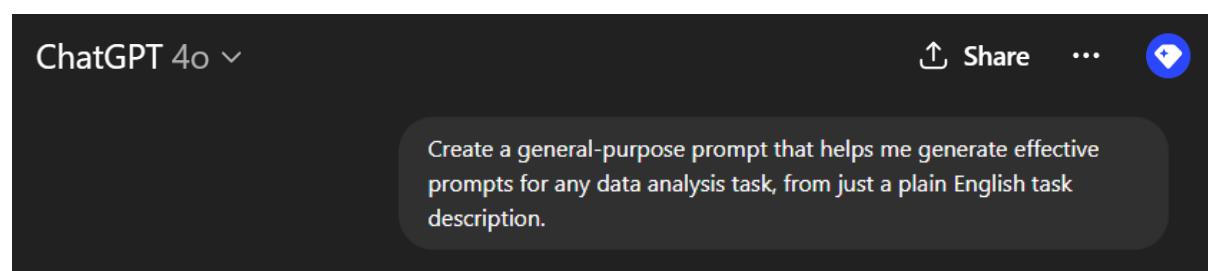
Positive Service Feedback:

1. "..."
2. "..."
3. "..."

Purpose: Shows auto-generation of audience-specific prompt variants.

Meta-Prompting for Prompt Automation

Prompt: Create a general-purpose prompt that helps me generate effective prompts for any data analysis task, from just a plain English task description.



ChatGPT Response:

Prompt: Generate a Structured Prompt for Data Analysis

I will give you a plain English description of a data analysis task. Your job is to turn it into a **well-structured, clear, and reusable prompt** that can be used with a language model or automation tool to perform the task consistently.

Instructions:

- "Interpret the task and extract the data type, goal, and expected output format."
- "Structure the prompt with:"
 - "A brief task overview"
 - "A clear list of step-by-step instructions"
 - "A consistent output format"
- "Use a neutral, precise, and task-specific tone (avoid unnecessary generalities)."
- "Make sure the prompt can be reused across different datasets of the same type."
- "Do not assume additional data unless explicitly mentioned."

Input:

Plain English task description:

"<insert user's description here>"

Output:

A fully-formed prompt ready to be used with a language model or automation tool.

Example Usage:

Input: "I want to analyze a list of sales transactions and find which products are most popular, which regions are underperforming, and generate a weekly trend summary."

Output Prompt:

You are analyzing a structured dataset of sales transactions. Your goal is to:

1. "Identify the top 5 most frequently sold products."
2. "Determine the bottom 3 performing regions based on total sales volume."
3. "Generate a weekly sales trend summary."

Assume the dataset includes fields like `product_name`, `region`, `date`, and `sales_amount`.

Output Format:

Top Products:

1. "Product A – X units"
2. "..."

Underperforming Regions:

1. "Region X – \$Y
..."

Weekly Sales Trend:

Week 1: \$Total

Week 2: \$Total

...

Purpose: Introduces meta-prompting, generating prompts about generating prompts.

Prompt Evaluation and Ranking

Prompt: Here are three versions of a prompt for summarizing customer reviews. Rate each one for clarity, completeness, and business utility on a scale of 1 to 5, and explain your reasoning.

(Paste three prompt versions)

ChatGPT Response:

Here is a detailed evaluation of each prompt using a 5-point scale for **Clarity**, **Completeness**, and **Business Utility**, along with reasoning.

1. For Internal Analysts (Data-Focused, Neutral)

Clarity: 5/5

The instructions are precise, clearly describing the task and expected tone. It avoids ambiguity and explicitly defines formatting.

Completeness: 5/5

All key components of a good data summary prompt are included—task scope, tone, grouping strategy, and format. It also avoids assumptions and subjective interpretation, which aligns well with an analyst's responsibilities.

Business Utility: 5/5

Perfectly suited for internal stakeholders who need structured data to inform decision-making. The format makes it easy to plug into dashboards or internal reports.

 **Overall Rating: 15/15**

An excellent prompt for objective, trend-focused internal analysis.

2. For Marketing Teams (Value-Focused, Messaging-Oriented)

Clarity: 4.5/5

Mostly clear, though slightly less structured than the analyst version. The phrase "possible language to reinforce or avoid" could be interpreted broadly and may benefit from an example.

Completeness: 4.5/5

Covers core elements like positive sentiment and brand risk, but could improve by explicitly asking the user to **relate insights to messaging tone or campaigns** (e.g., "which themes align with current messaging?").

Business Utility: 5/5

Very useful for content creators and brand managers. Identifying what resonates (or backfires) with customers directly supports positioning and copy refinement.

 **Overall Rating: 14/15**

Strong and adaptable, with a slight room for tighter framing of expectations.

👉 3. For Customer Support Leads (Action-Oriented, Service-Driven)

Clarity: 5/5

Clear instructions with a logical, easy-to-follow structure. The phrase "support process improvements or FAQs" gives useful context.

Completeness: 5/5

It includes both complaint trends and praise, focuses on continuous improvement, and even hints at how the feedback can be operationalized (e.g., FAQs, resolution workflows).

Business Utility: 5/5

Excellent for support teams. It not only identifies issues but guides actionable next steps to improve satisfaction and reduce support costs.

✅ Overall Rating: 15/15

Highly targeted and impactful for CX and support operations.

📊 Summary Comparison Table

Prompt Version	Clarity	Completeness	Business Utility	Total
Internal Analysts	5	5	5	15
Marketing Teams	4.5	4.5	5	14
Customer Support Leads	5	5	5	15

Purpose: Adds evaluation and validation, a key part of the auto-prompting loop.



Summary

By the end of this demo, learners will have:

- Auto-generated prompts from plain task goals
- Refined them with tone and output requirements
- Created variants for different departments
- Generated prompts from examples (few-shot inference)
- Used meta-prompts to automate future prompt design
- Evaluated and ranked prompt quality

Conclusion

This demo shows how auto-prompting streamlines prompt creation from task descriptions or examples. Learners gain practical skills to generate, refine, and adapt prompts for real-world business tasks, making prompt-driven systems more scalable and efficient.